

BLM2041 Signals and Systems

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BLM2041 Signals and Systems

Sampling & Aliasing

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LECTURE OBJECTIVES

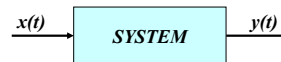
- SAMPLING can cause ALIASING
 - Sampling Theorem
 - Sampling Rate > 2(Highest Frequency)
- Spectrum for digital signals, $x[n]$
 - Normalized Frequency

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi \ell$$

↑
ALIASING

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SYSTEMS Process Signals



- PROCESSING GOALS:
 - Change $x(t)$ into $y(t)$
 - For example, more BASS
 - Improve $x(t)$,
 - e.g., image deblurring
 - Extract information from $x(t)$

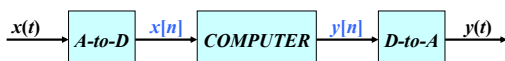
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System IMPLEMENTATION

- ANALOG/ELECTRONIC:
 - Circuits: resistors, capacitors, op-amps



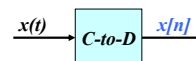
- DIGITAL/MICROPROCESSOR
 - Convert $x(t)$ to numbers stored in memory



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SAMPLING $x(t)$

- SAMPLING PROCESS
 - Convert $x(t)$ to numbers $x[n]$
 - “ n ” is an integer;
 - $x[n]$ is a sequence of values
 - Think of “ n ” as the storage address in memory
- UNIFORM SAMPLING at $t = nT_s$
 - IDEAL: $x[n] = x(nT_s)$



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SAMPLING RATE, f_s

- SAMPLING RATE (f_s)

– $f_s = 1/T_s$

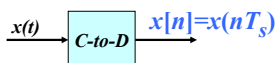
- NUMBER of SAMPLES PER SECOND

– $T_s = 125 \text{ microsec} \rightarrow f_s = 8000 \text{ samples/sec}$

– UNITS ARE HERTZ: 8000 Hz

- UNIFORM SAMPLING at $t = nT_s = n/f_s$

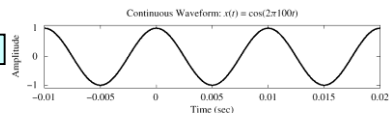
– IDEAL: $x[n] = x(nT_s) = x(n/f_s)$



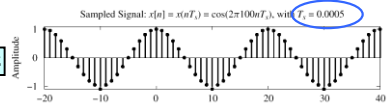
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SAMPLING RATE, f_s

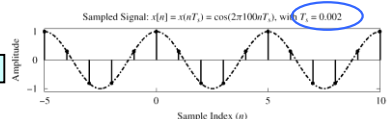
$f = 100\text{Hz}$



$f_s = 2 \text{ kHz}$



$f_s = 500\text{Hz}$



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SAMPLING THEOREM

- HOW OFTEN ?

– DEPENDS on FREQUENCY of SINUSOID

– ANSWERED by SHANNON/NYQUIST Theorem

– ALSO DEPENDS on “RECONSTRUCTION”

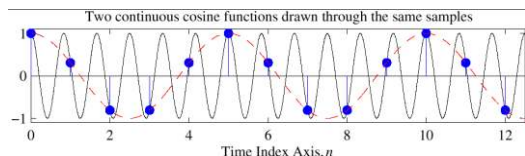
Shannon Sampling Theorem

A continuous-time signal $x(t)$ with frequencies no higher than $f_{\max}$ can be reconstructed exactly from its samples $x[n] = x(nT_s)$, if the samples are taken at a rate $f_s = 1/T_s$ that is greater than $2f_{\max}$.

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Reconstruction? Which One?

Given the samples, draw a sinusoid through the values



$x[n] = \cos(0.4\pi n)$

When n is an integer

$\cos(0.4\pi n) = \cos(2.4\pi n)$

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STORING DIGITAL SOUND

- $x[n]$ is a SAMPLED SINUSOID

– A list of numbers stored in memory

- EXAMPLE: audio CD

• CD rate is 44,100 samples per second

– 16-bit samples

– Stereo uses 2 channels

• Number of bytes for 1 minute is

– $2 \times (16/8) \times 60 \times 44100 = 10.584 \text{ Mbytes}$

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DISCRETE-TIME SINUSOID

- Change $x(t)$ into $x[n]$

- DERIVATION

$x(t) = A \cos(\omega t + \phi)$

$x[n] = x(nT_s) = A \cos(\omega nT_s + \phi)$

$x[n] = A \cos((\omega T_s)n + \phi)$

$x[n] = A \cos(\hat{\omega} n + \phi)$

$\hat{\omega} = \omega T_s = \frac{\omega}{f_s}$

DEFINE DIGITAL FREQUENCY

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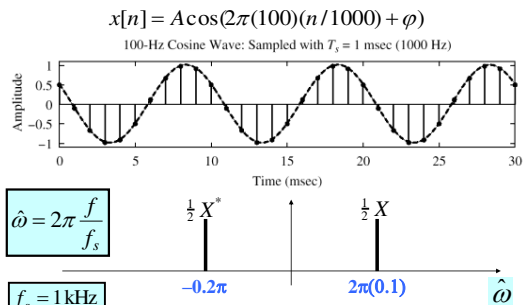
DIGITAL FREQUENCY

- Digital frequency $\hat{\omega}$ VARIES from 0 to 2π , as f varies from 0 to the sampling frequency
- UNITS are radians, not rad/sec
 - DIGITAL FREQUENCY is NORMALIZED

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s}$$

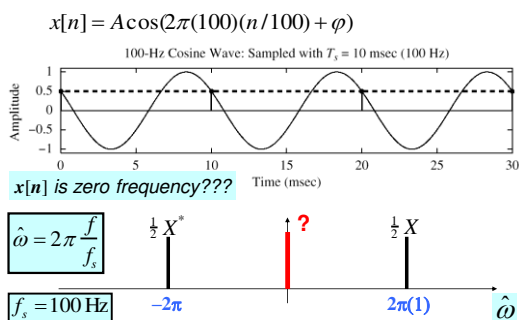
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SPECTRUM (DIGITAL)



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SPECTRUM (DIGITAL) ???



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The REST of the STORY

- Spectrum of $x[n]$ has more than one line for each complex exponential
 - Called ALIASING
 - MANY SPECTRAL LINES
- SPECTRUM is PERIODIC with period = 2π
 - Because

$$A \cos(\hat{\omega}n + \phi) = A \cos((\hat{\omega} + 2\pi)n + \phi)$$

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ALIASING DERIVATION

- Other Frequencies give the same $\hat{\omega}$

$$\begin{aligned} x_1(t) &= \cos(400\pi t) \quad \text{sampled at } f_s = 1000 \text{ Hz} \\ x_1[n] &= \cos(400\pi \frac{n}{1000}) = \cos(0.4\pi n) \\ x_2(t) &= \cos(2400\pi t) \quad \text{sampled at } f_s = 1000 \text{ Hz} \\ x_2[n] &= \cos(2400\pi \frac{n}{1000}) = \cos(2.4\pi n) \\ x_2[n] &= \cos(2.4\pi n) = \cos(0.4\pi n + 2\pi n) = \cos(0.4\pi n) \\ \Rightarrow x_2[n] &= x_1[n] \quad 2400\pi - 400\pi = 2\pi(1000) \end{aligned}$$

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ALIASING DERIVATION-2

- Other Frequencies give the same $\hat{\omega}$

If $x(t) = A \cos(2\pi(f + \ell f_s)t + \phi)$ $t \leftarrow \frac{n}{f_s}$

and we want $x[n] = A \cos(\hat{\omega}n + \phi)$

then: $\hat{\omega} = \frac{2\pi(f + \ell f_s)}{f_s} = \frac{2\pi f}{f_s} + \frac{2\pi \ell f_s}{f_s}$

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi \ell$$

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ALIASING CONCLUSIONS

- ADDING f_s or $2f_s$ or $-f_s$ to the FREQ of $x(t)$ gives exactly the same $x[n]$
 - The samples, $x[n] = x(n/f_s)$ are **EXACTLY THE SAME VALUES**
- GIVEN $x[n]$, WE CANNOT DISTINGUISH f_o FROM $(f_o + f_s)$ or $(f_o + 2f_s)$

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NORMALIZED FREQUENCY

- DIGITAL FREQUENCY

Normalized Radian Frequency

$$\hat{\omega} = \omega T_s = \frac{2\pi f}{f_s} + 2\pi\ell$$

Normalized Cyclic Frequency

$$\hat{f} = \hat{\omega}/(2\pi) = f T_s = f/f_s$$

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SPECTRUM for $x[n]$

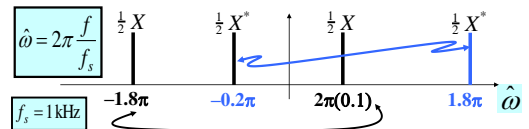
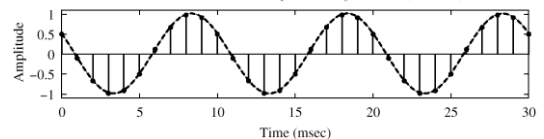
- PLOT versus NORMALIZED FREQUENCY
- INCLUDE **ALL** SPECTRUM LINES
 - **ALIASES**
 - ADD MULTIPLES of 2π
 - SUBTRACT MULTIPLES of 2π
 - **FOLDED ALIASES**
 - ALIASES of NEGATIVE FREQS

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SPECTRUM (MORE LINES)

$$x[n] = A \cos(2\pi(100)(n/1000) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 1$ msec (1000 Hz)

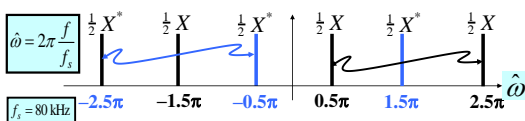
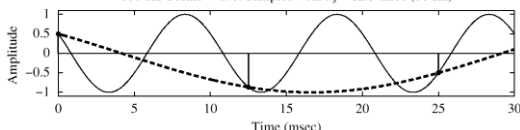


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SPECTRUM (ALIASING CASE)

$$x[n] = A \cos(2\pi(100)(n/80) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 12.5$ msec (80 Hz)

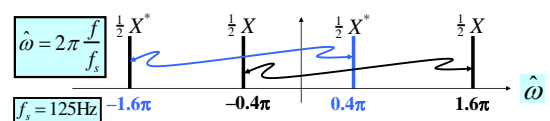
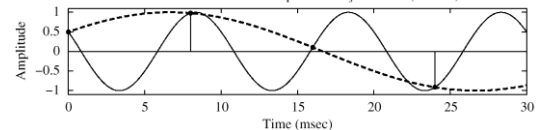


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SPECTRUM (FOLDING CASE)

$$x[n] = A \cos(2\pi(100)(n/125) + \varphi)$$

100-Hz Cosine Wave: Sampled with $T_s = 8$ msec (125 Hz)



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